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Space-charge compensation measurements in electron cyclotron resonance ion source low energy beam transport lines with a retarding field analyzer^{a)}

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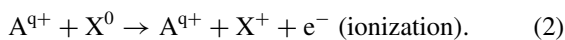
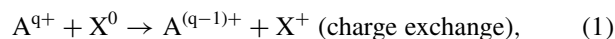
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In this paper we describe the first systematic measurement of beam neutralization (space charge compensation) in the ECR low energy transport line with a retarding field analyzer, which can be used to measure the potential of the beam. Expected trends for the space charge compensation levels such as increase with residual gas pressure, beam current, and beam density could be observed. However, the overall levels of neutralization are consistently low (<60%). The results and the processes involved for neutralizing ion beams are discussed for conditions typical for ECR injector beam lines. The results are compared to a simple theoretical beam plasma model as well as simulations. © 2014 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4854315>]

I. INTRODUCTION

With the dramatic performance increase of state-of-the-art electron cyclotron resonance ion sources (ECRIS) and the demand for ever more intense beams for next generation heavy ion and rare isotope facilities such as the facility for rare isotope beams (FRIB), total extracted and transported beam current of 5–10 emA before the charge state selection and a few 100 to several hundreds of eμA after the charge selection are routinely used. At these high intensities space charge and space charge compensation have become important factors for realistic beam simulations and accelerator design. Space charge compensation (beam neutralization) occurs when the beam interacts with the residual gas in the beam line and creates slow secondary ions and electrons. The positive ions of the beam interact with the gas molecules mainly through two processes; charge exchange and ionization:



Only the second process creates the desired secondary electrons that can be trapped in the beam and reduce the beam potential. The slow (secondary) ions are expelled by the beam potential and can be used to quantify the neutralization level by measuring their energy distribution (e.g., with a retarding field analyzer (RFA)). Typically, ECR ion sources and low energy beam transport lines are designed to operate at very low residual gas pressures between 10^{-6} Torr and 10^{-9} Torr to minimize losses of highly charged ions due to charge exchange. Therefore the production rate of secondary particles is rather low. On the other hand, ECR injectors for heavy ion beam facilities are typically operated in continuous mode and

one might assume that space charge compensation should be high in these beam lines as long as loss terms for the trapped electrons are low. Contrary to this simple hand waving argument, it is often found experimentally that the focusing strengths of optical elements need to be increased with increasing ion beam current, but very few measurements exist to evaluate the space charge compensation for high charge state ion injectors.¹

This paper describes the experimental setup and presents neutralization factor measurements on the SuSI (Superconducting Source for Ions) injection line at the Coupled Cyclotron Facility (CCF) at Michigan State University. In addition, a model adapted from earlier works by Gabovich and Soloshenko^{2,3} is compared to the measurements (Sec. IV).

II. EXPERIMENTAL SETUP

A. Retarding field analyzer

The RFA⁴ consists of three parallel high transparent copper meshes (combined transmission ~73%) housed in a grounded box with a set of two 10 mm diameter apertures for collimation of the secondary ions upon entering the detector (see Fig. 1). Mesh 1 is grounded at all times and helps providing a uniform retarding field together with mesh 2 (biased between −100 V and +200 V). Mesh 3 is typically set to −150 V and acts as an electron repeller, both to shield electrons from entering the detector and suppress secondary electrons produced upon impact of the measured ions on the collector (Faraday Cup).

The RFA is mounted perpendicular to the beam to measure the energy distribution of the expelled secondary ions. For the measurements presented at the SuSI injector line the RFA was mounted after the analyzing magnet in the collimator channel of the CCF injection line in one of the diagnostic boxes. In addition to the RFA, a Faraday Cup, a beam viewer and a four-jaw collimator are mounted in the box. In

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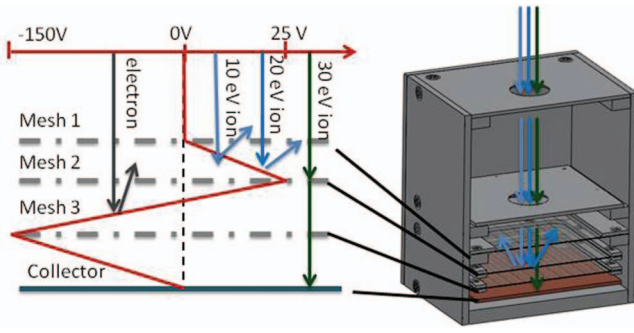


FIG. 1. 3D CAD model of the RFA is shown on the right. The picture on the left depicts the working principle of the RFA. In the picture the retarding grid is biased at 25 V, ions with less than 25 eV energy will be reflected by the retarding field. Two 10 mm diameter apertures collimate the incoming secondary ions.

order to systematically vary the pressure during the measurements, a gas valve and a pressure gauge were installed as well.

Due to the finite size of the wires and distances between wires, the equipotential lines close to the retarding grid are not straight and influence the measured spectrum as well as the transmission through the wires. These effects were carefully taken into account in the analysis of the spectra. A detail description of these effects and how they were considered in the data analyses can be found in Refs. 4 and 5.

III. ANALYSES

In the approximation of a round beam, the potential difference between the center and the edge of the beam is given by

$$\Delta\phi_{meas} = \frac{I \cdot (1 - f)}{4\pi \cdot \epsilon_0 \cdot \beta c} = \Delta\phi_I(1 - f), \quad (3)$$

where I is the total beam current, ϵ_0 is the vacuum permittivity, βc is the velocity of the beam, $\Delta\phi_I$ is the non-neutralized beam potential, and f is the neutralization factor between 0 and 1.

The potential difference $\Delta\phi_{meas}$ can be obtained from the RFA spectrum by taking the derivative dI/dV which yields the energy distribution of the ions. A typical spectrum measured at the SuSI injection line is shown in Fig. 2 together with its first derivative.

In order to obtain the neutralization factor f the measured $\Delta\phi_{meas}$ is compared with the calculated non-neutralized beam potential (see Eq. (3)). In addition to the beam potential, the saturation current in the detector (measured with the retarding grid 2 potential set to 0 V) is compared with the theoretically expected saturation current for a given beam current and residual gas density.

The saturation current is calculated by considering the total production of ions per unit time using the beam current I_b ,

$$\frac{dn_i}{dt} = lI_b \sum n_g \cdot \sigma_{i,g}; \quad I_b = qer_b^2 \pi n_b v_b, \quad (4)$$

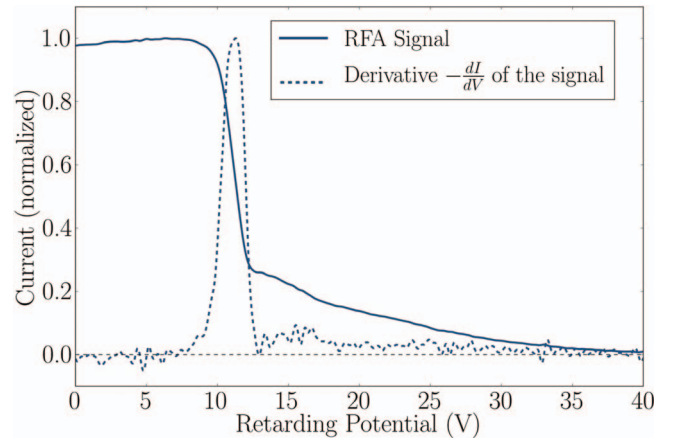


FIG. 2. Typical RFA signal measured at the SuSI injector line.

and the solid angle into the detector,

$$I_{sat} = \frac{r_a \cdot \arctan \frac{r_a}{d} I_b \cdot T \cdot \sum n_g \cdot \sigma_{i,g}}{2 \cdot q \cdot e} \approx \frac{r_a^2 \cdot I_b \cdot T \cdot \sum n_g \cdot \sigma_{i,g}}{2 \cdot d \cdot q \cdot e}, \quad (5)$$

where r_a is the entrance aperture of the detector, d is the distance of the detector, I_b is the electrical ion beam current, q is the primary ion charge state, $\sigma_{i,g}$ is the total secondary ion production cross section, and n_g is the residual gas density.

In general good agreement was found between the measured and calculated saturation currents, which adds confidence that the corrections for the secondary ion transmission into the detectors were made correctly.⁴

IV. MEASUREMENTS

As an example, Fig. 3 shows the measured saturation currents and derived neutralization factors of a 200 μA Ar^{8+} beam and a 700 μA O^{6+} beam in dependence of the beam line pressure. The beam line pressure was varied between 1×10^{-6} Torr and 1×10^{-5} Torr. In addition, the theoretical prediction applying the 1D beam plasma model (see Sec. V) derived by Soloshenko and Gabovich^{2,3} are also indicated in the graph. The shaded areas capture the uncertainties of the cross sections ($\pm 25\%$) and applicable measurement uncertainties. The model fits the data reasonably well within the error bars. As expected the neutralization level increases with increasing pressure. In general, the measured (and calculated) neutralization factors even at the relative high beam line pressure of 1×10^{-5} Torr are rather low. On first glance this is a surprising result since measurements in the LEBT section of high intensity sources such as the Low Energy Demonstration Experiment (LEDA) injector source showed 95%–90% neutralization factors for beam currents of 130 mA protons⁶ and neutralization factors of 80% for beam currents between 2 and 10 mA protons.^{4,5} However, these results are consistent with the presented measurements when the different beam current intensities are taken into account. This will be further discussed in Secs. V and VI.

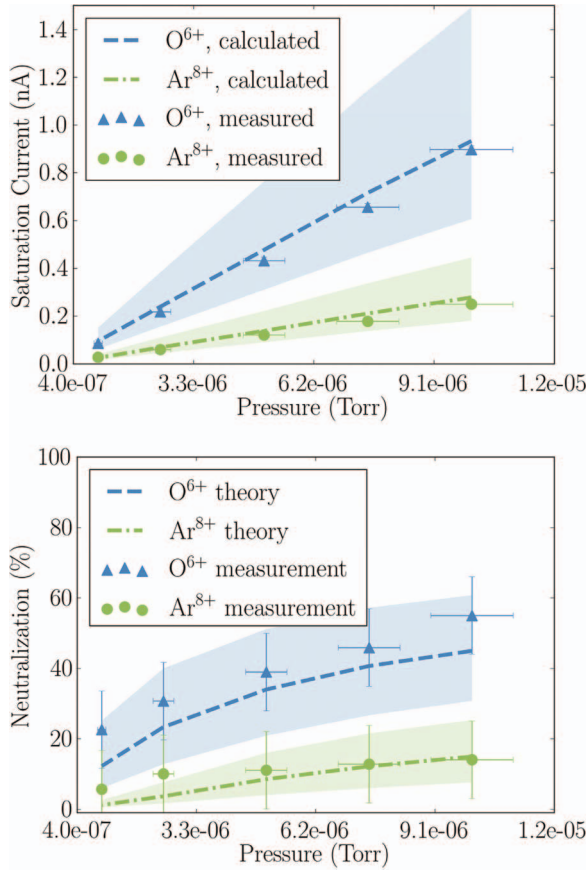


FIG. 3. Saturation currents (top) and neutralization factors (bottom) derived from the spectrum for a 200 e μ A Ar⁸⁺ and a 700 e μ A O⁶⁺ beam as a function of the beam line pressure. The dashed lines indicate the model prediction, the shaded areas show the to the uncertainties of the prediction due to absolute value of the cross sections.

V. DISCUSSION

In Ref. 3, Soloshenko derives a 1D plasma model for a steady state continuous mode (round) beam, based on an energy balance of the secondary electrons. Electrons captured inside the beam may gain enough energy to leave through Coulomb collisions with the beam ions themselves. This energy gain must be equal to the energy taken out of the system by the leaving electrons. In addition, the model assumes quasi-neutrality of the beam plasma (the number of electrons must be equal to the beam ions and secondary ions) and requires the number of secondary electrons and ions created in the beam to be equal to the number of electrons and ions leaving the beam per unit time, respectively. Following the derivation in Refs. 2 and 3, the potential difference between the beam center and beam edge ($\Delta\phi = \Delta\phi_{\text{meas}} = \phi_{\text{center}} - \phi_{\text{edge}}$) can be expressed in SI units as

$$(\Delta\phi_{\text{meas}})^2 = 3\mathcal{L} \left(\frac{M}{m_e} \right) \left(\frac{\varphi_i}{V_0} \right) \frac{n_b q e^2}{(4\pi\epsilon_0)^2} \left(\frac{q}{n_g \sigma_e} + \frac{v_b \sigma_i r_b}{2v_i \sigma_e} \right), \quad (6)$$

where $\mathcal{L}$ is the Coulomb logarithm, M is the beam ion mass, φ_i is the gas ionization potential, V_0 is the source voltage, v_i is the average velocity of the secondary ions created in the interaction between the beam and the residual gas, v_b is the ion

beam velocity, n_b is the ion beam density, n_g is the residual gas density, σ_i is the total ion production cross-section, σ_e is the electron production cross-section, and r_b is the ion beam radius. The average secondary ion velocity v_i depends on the potential well depth of the compensated beam. The dependence goes as the square root of the beam potential, though making it a fourth root contribution to the calculated. Thus choosing an ion temperature close to the expected beam potential gives good results and even a rather large variation of it does not change the result significantly.⁴ For low pressures we can expect a decrease in $\Delta\phi$ with increasing pressure. For high pressure, $\Delta\phi$ reaches a minimum and becomes essentially independent with pressure (beam is fully neutralized). Notably, the reduced beam potential also depends on the ion beam radius and density. For ECR ion sources the residual gas pressure is typically low and the ion beam current after the analyzing magnet is below 1 mA. At these low currents, the beam potential is low (a few 10 s of volt) and the production rate of the secondary electron is not high enough to compensate the losses of the electrons through collisions with the beam ions. Consequently the neutralization factors are low. For the data presented here, the formalism was extended to lower neutralization factors by replacing the quasineutrality condition with a partially neutralized plasma condition for calculating the electron density in the beam.^{2,3}

VI. IMPLEMENTATION OF THE MODEL INTO BEAM TRANSPORT SIMULATION

Using this formalism a simple neutralization model can be added to the beam line simulation. The neutralization factor for a given pressure at the specific beam line location is calculated by applying the formula presented above at each step of the simulation. The calculated neutralization factor is then used to weigh the ion current of each species for the next step of the calculation. In Fig. 4 the neutralization level from the source exit through the analyzing section of the SuSI injector beam line is plotted for a multispecies oxygen beam. Since SuSI utilizes an electrostatic einzel lens as first focusing element, full space charge is present after the extraction system. In the drift before the analyzing magnet, where the beam current is still high (due to the many species in the beam) and the beam optics was designed to

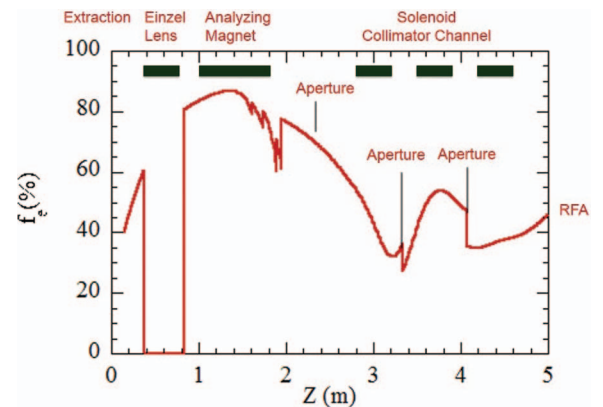


FIG. 4. Neutralization levels along the beam line in the SuSI LEBT.

keep the beam radius large before the analyzing magnet the neutralization factor can reach very high values. The discontinuities in the neutralization factor inside the magnetic bend are due to the change in particle current as one oxygen charge state after the other is lost in the analyzing magnet. Although the model oversimplifies the neutralization processes along the beam line, it demonstrates how the neutralization factor can vary as the beam is transported through the low energy beam transport system. The neutralization values extracted from the beam line simulation of 50% at the location of the RFA for the $235\text{ }\mu\text{A O}^{3+}$ beam agrees well with the measured neutralization factor of 57% using the RFA.

VII. SUMMARY

The level of space charge neutralization was measured after the analyzing magnet of the SuSI ECR ion source at Michigan State University. The measurements suggest overall low neutralization factors (0%–60%) at this location. The

factors derived compare reasonably well with the extended theoretical model introduced by Gabovich and Soloshenko. The neutralization level along the beam line depends on the pressure, the beam diameter, and the beam intensity. Therefore the neutralization factor can vary considerably as the beam is transported through the low energy beam transport system.

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